

A Note on the Value Theorems of Calculus

The Intermediate Value Property, and the Mean Value Theorems

Before we begin

Calculus has a handful of theorems that feel almost too simple to be useful — and the Value Theorems are the champions of that club. They look like cozy geometric observations (“a continuous curve must cross every intermediate height”; “a smooth curve must have a tangent parallel to its chord somewhere”), and yet, quietly, they do a staggering amount of the heavy lifting in real analysis: bounding functions, proving inequalities, counting roots, comparing growth rates, and even powering L’Hôpital’s rule.

This note is a walk through four siblings in this family — the **Intermediate Value Theorem (IVT)** and the three Mean Value Theorems of **Rolle**, **Lagrange**, and **Cauchy**. Along the way, we’ll ask some questions that standard textbooks often skip: does a function *need* to be continuous to have the intermediate-value property? Which discontinuities are compatible with it? And what, exactly, is the strange and beautiful theorem of Darboux that says “every derivative, continuous or not, has the intermediate-value property”?

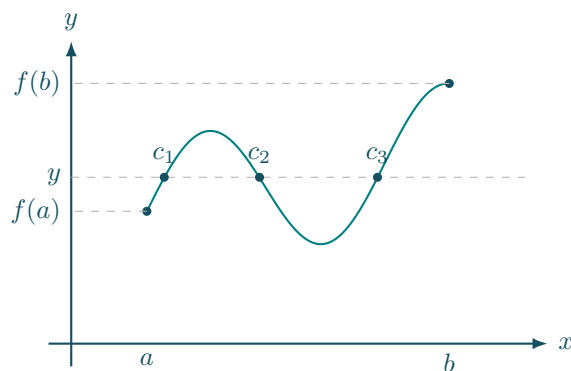
Contents

1	The Intermediate Value Theorem — the existence engine	1
1.1	The intermediate-value property, and when continuity fails	3
1.2	Darboux’s Theorem — every derivative has the IVP	4
2	Rolle’s Theorem — the quiet ancestor	5
3	Lagrange’s Mean Value Theorem — the workhorse	7
4	Cauchy’s Mean Value Theorem — the comparator	8
5	A cheat sheet	10
6	Exercises	11

1 The Intermediate Value Theorem — the existence engine

Theorem (Intermediate Value Theorem (IVT)). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If y is any value between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.*

Motivation. Imagine drawing a continuous curve from altitude $f(a)$ to altitude $f(b)$ without lifting your pen. If you start at 2 metres and end at 5 metres, at some moment you were at exactly 3 metres, exactly π metres, exactly any height in between.



Intuition. The IVT is really a statement about *connectedness*: a continuous image of a connected set is connected. In $\mathbb{R}$, the connected sets are intervals, and intervals have no gaps. So a continuous f sweeps through an interval of y -values with no holes. The most-used special case: if $f(a)$ and $f(b)$ have opposite signs, f has a root in (a, b) .

Which theorem to reach for?. Reach for **IVT** when you need to prove that *something exists* — a root, a fixed point, a matching pair — and your function is merely continuous (not necessarily differentiable). The classic tell: an equation where you can spot a sign change, or a clever auxiliary function $h(x) = f(x) - g(x)$ whose zeros are what you want.

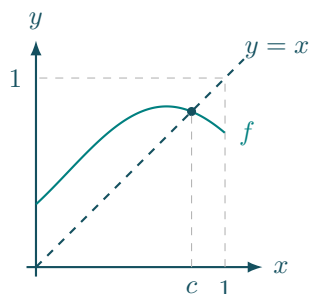
Example 1.1: A root where none is visible. Prove that $x^3 - 4x + 1 = 0$ has three real roots.

Solution. Let $f(x) = x^3 - 4x + 1$. Compute: $f(-3) = -14$, $f(0) = 1$, $f(1) = -2$, $f(3) = 16$. So f changes sign on $[-3, 0]$, on $[0, 1]$, and on $[1, 3]$. By IVT, there is a root in each of these three disjoint intervals. A cubic has at most three real roots, so these are all of them. $\square$

We didn't solve the cubic; we *located* its roots using four arithmetic evaluations and the IVT.

Example 1.2: A fixed point in disguise. Let $f : [0, 1] \rightarrow [0, 1]$ be continuous. Prove that f has a fixed point.

Solution. Consider $g(x) = f(x) - x$. Then g is continuous, $g(0) = f(0) \geq 0$, and $g(1) = f(1) - 1 \leq 0$. If either endpoint gives equality we are done; otherwise $g(0) > 0 > g(1)$, and by IVT there is $c \in (0, 1)$ with $g(c) = 0$, i.e. $f(c) = c$. $\square$



Fixed-point problems are IVT problems in disguise. This one-dimensional Brouwer theorem is the toy model for deep fixed-point results in topology and economics.

Example 1.3: Antipodal temperatures. Show that at any moment there exist two diametrically opposite points on the Earth's equator having exactly the same temperature.

Solution. Let $T(\theta)$ be the (continuous) temperature at angle $\theta \in [0, 2\pi]$. Define $h(\theta) = T(\theta) - T(\theta + \pi)$ on $[0, \pi]$. Then $h(\pi) = T(\pi) - T(2\pi) = T(\pi) - T(0) = -h(0)$, so $h(0)$ and $h(\pi)$ have opposite signs (or one is zero). IVT gives θ_0 with $h(\theta_0) = 0$, i.e. $T(\theta_0) = T(\theta_0 + \pi)$. $\square$

The trick — *subtract the function from its shifted version to engineer a sign change* — is the one-dimensional Borsuk–Ulam theorem.

Example 1.4: A meeting of two hikers. Two hikers start at opposite ends of a rugged mountain trail of length L , at the same moment, walking toward each other. Hiker A’s position at time t is $a(t)$, continuous, with $a(0) = 0, a(T) = L$. Hiker B’s is $b(t)$, continuous, with $b(0) = L, b(T) = 0$. Prove they meet at some moment.

Solution. Let $h(t) = a(t) - b(t)$. This is continuous, $h(0) = -L < 0$, and $h(T) = L > 0$. By IVT, some $t_0 \in (0, T)$ satisfies $h(t_0) = 0$, so $a(t_0) = b(t_0)$. $\square$

“Meeting” problems — two clocks showing the same time, two monks ascending and descending a path on successive days — are all “sign change of a difference” problems. Same IVT, different costumes.

Example 1.5: Polynomials of odd degree. Prove that every polynomial of odd degree with real coefficients has at least one real root.

Solution. Let $p(x) = a_n x^n + \cdots + a_0$ with n odd and $a_n \neq 0$. For $|x|$ large, the leading term dominates: if $a_n > 0$, then $p(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (because n is odd), and vice versa if $a_n < 0$. Either way, p takes both positive and negative values. Since p is continuous, IVT produces a root. $\square$

This is the cleanest proof of a fundamental-sounding fact — one that doesn’t require the Fundamental Theorem of Algebra. Odd degree forces opposite-sign limits at infinity, and IVT does the rest.

1.1 The intermediate-value property, and when continuity fails

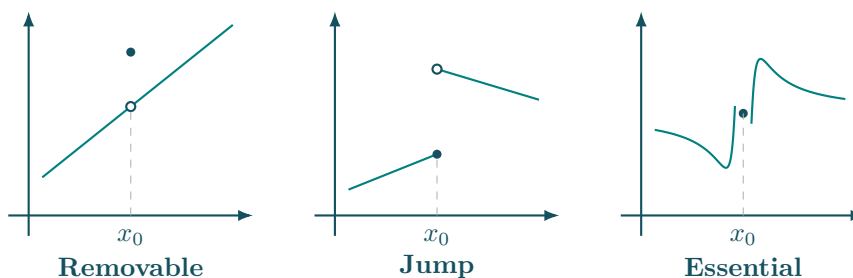
The IVT says *continuous* $\Rightarrow$ *intermediate-value property*. But is the converse true? If a function has the intermediate-value property (IVP for short — it hits every value between any two of its values), must it be continuous?

Surprisingly, no. And the exceptions are worth understanding.

Three types of discontinuity

Let f be defined near x_0 but discontinuous there. Write $L^- = \lim_{x \rightarrow x_0^-} f(x)$ and $L^+ = \lim_{x \rightarrow x_0^+} f(x)$ (when they exist). The classification is:

- Removable discontinuity:** Both one-sided limits exist and are equal: $L^- = L^+ = L$, but $L \neq f(x_0)$ (or $f(x_0)$ is undefined). The graph has a single “hole” that could be plugged by redefining one value.
- Jump discontinuity:** Both one-sided limits exist but $L^- \neq L^+$. The graph leaps from one level to another.
- Essential (oscillatory) discontinuity:** At least one of the one-sided limits fails to exist — typically because the function oscillates wildly, like $\sin(1/x)$ near 0.



Removable and jump discontinuities share a feature: both one-sided limits exist. Call these *first-kind* (or *simple*) discontinuities. Essential discontinuities are *second-kind* — at least one one-sided limit fails.

IVP forbids simple discontinuities

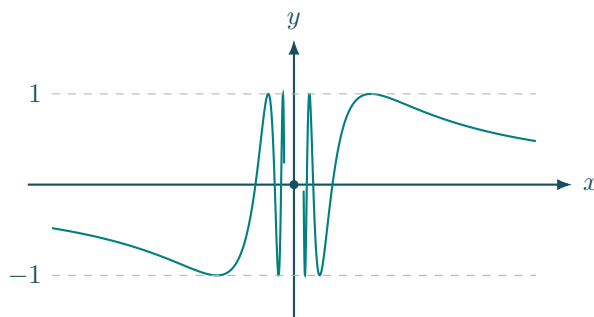
Theorem (IVP rules out simple discontinuities). *If $f : I \rightarrow \mathbb{R}$ has the intermediate-value property on an interval I , then f has no removable and no jump discontinuities on I . The only discontinuities it can have are essential.*

Why this is true. Suppose f had a jump at x_0 : say $L^- < L^+$ and $f(x_0)$ sits somewhere (or is a third value). Pick any y strictly between L^- and $\min(L^+, f(x_0))$, but different from $f(x_0)$. By the one-sided limits, in a punctured neighbourhood of x_0 the function f either stays near L^- (on the left) or near L^+ (on the right), so it misses y entirely except possibly at x_0 — and we chose $y \neq f(x_0)$. Then f cannot hit y on any small interval straddling x_0 , contradicting IVP. A similar (simpler) argument handles removable discontinuities. The only way a discontinuity can survive IVP is for f to oscillate so wildly near x_0 that it *does* keep crossing every nearby value — which is exactly what an essential discontinuity looks like.

The canonical example. Define

$$f(x) = \begin{cases} \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then f has an essential discontinuity at 0 (the limit $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist — f oscillates between -1 and $+1$ infinitely often in every neighbourhood of 0). Yet f has the intermediate-value property on every interval, including those containing 0: on any interval around 0, f takes every value in $[-1, 1]$ (infinitely often, in fact), and on intervals away from 0 it's continuous.



Moral: IVP is a *strictly weaker* condition than continuity. Continuous functions have IVP, but IVP allows one kind of bad behaviour — wild oscillation — that continuity forbids.

1.2 Darboux's Theorem — every derivative has the IVP

Here is one of the prettiest and most surprising results in elementary analysis. A function f need not be differentiable; but if it *is* differentiable, the derivative f' — even if f' is itself discontinuous — must have the intermediate-value property.

Theorem (Darboux's Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable on $[a, b]$. Then f' has the intermediate-value property on $[a, b]$: for any y strictly between $f'(a)$ and $f'(b)$, there exists $c \in (a, b)$ with $f'(c) = y$.*

Why this is surprising. Derivatives can be discontinuous. The standard example,

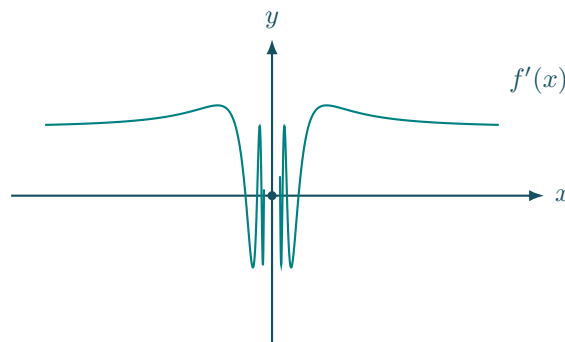
$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases} \quad f'(0) = 0, \quad f'(x) = 2x \sin(1/x) - \cos(1/x) \text{ for } x \neq 0,$$

has a derivative that oscillates wildly near 0 (because of the $-\cos(1/x)$ term) — no limit at 0, so f' has an essential discontinuity there. By our previous theorem, f' *had* to have an essential discontinuity if it was going to be discontinuous at all: Darboux’s theorem tells us f' has the IVP, so no jump or removable discontinuities are allowed.

Sketch of proof. We prove the sign-change special case — the full IVP reduces to this by subtracting a linear function. Suppose $f'(a) < 0 < f'(b)$; we show $f'(c) = 0$ for some $c \in (a, b)$.

Since f is continuous on $[a, b]$, it attains its minimum at some $c \in [a, b]$. We rule out the endpoints: near a , since $f'(a) < 0$, the definition of the derivative gives $f(x) < f(a)$ for x slightly greater than a , so a is not the minimum. Similarly, $f'(b) > 0$ means $f(x) < f(b)$ for x slightly less than b , ruling out b . So $c \in (a, b)$ is an interior minimum, and $f'(c) = 0$ by Fermat’s theorem. $\square$

What Darboux really says. Imagine you built a function F by integration: $F(x) = \int_a^x g(t) dt$. Then $F' = g$ (where g is continuous). Darboux tells you that *not every* bounded function g can be the derivative of *something*: only functions with the IVP are candidates. In this sense, “being a derivative” is a restrictive property, even when discontinuities are allowed.

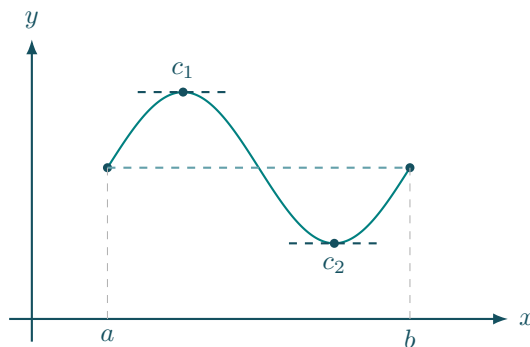


The derivative of $x^2 \sin(1/x)$ (extended by 0 at 0): essentially discontinuous at 0, yet satisfies the IVP.

2 Rolle’s Theorem — the quiet ancestor

Theorem (Rolle’s Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.*

Motivation. Picture a hiker who starts and ends a trek at exactly the same altitude. No matter how wild the route in between, there must be at least one instant where the hiker is moving neither up nor down — a moment of level ground. That moment is c , and “level ground” is $f'(c) = 0$.



Intuition. Rolle is a statement about *extrema*. On a closed interval a continuous function attains its max and min; if the endpoints agree, at least one extremum lives strictly inside, and at an interior extremum of a differentiable function, the derivative vanishes.

Which theorem to reach for?. Reach for **Rolle** when you see — or can *engineer* — two points where a function takes equal values, and you want to extract a zero of its derivative. Classic tells: a question about how many roots f or f' can have; a weighted sum of coefficients equal to zero (antiderivative trick); a request to show some derivative expression vanishes.

Example 2.1: Counting roots through derivatives. Show that $x^5 + x^3 + x + 1 = 0$ has exactly one real root.

Solution. Let $f(x) = x^5 + x^3 + x + 1$. Since $f(-1) = -2 < 0$ and $f(0) = 1 > 0$, IVT gives at least one root. For uniqueness, suppose there were two roots $\alpha < \beta$. Then $f(\alpha) = f(\beta) = 0$, so by Rolle some $c \in (\alpha, \beta)$ satisfies $f'(c) = 0$. But $f'(x) = 5x^4 + 3x^2 + 1 \geq 1 > 0$ always — contradiction. $\square$

IVT gives existence, Rolle gives uniqueness. Any time you want to show “exactly one root,” think IVT+Rolle.

Example 2.2: Roots of a derivative you can’t write down. Let $f(x) = (x-1)(x-2)(x-3)(x-4)$. Show f' has three real roots, one in each of $(1, 2)$, $(2, 3)$, $(3, 4)$.

Solution. $f(1) = f(2) = 0$ gives $c_1 \in (1, 2)$ with $f'(c_1) = 0$ by Rolle; similarly $c_2 \in (2, 3)$ and $c_3 \in (3, 4)$. Since f' is a cubic with at most three roots, these are all. $\square$

Rolle lets us locate roots of f' without computing them. Between consecutive roots of f sits a root of f' — the seed of the Gauss–Lucas theorem.

Example 2.3: An engineered equality. If $\frac{a_0}{1} + \frac{a_1}{2} + \cdots + \frac{a_n}{n+1} = 0$, show $a_0 + a_1x + \cdots + a_nx^n = 0$ has a root in $(0, 1)$.

Solution. Let $F(x) = a_0x + \frac{a_1}{2}x^2 + \cdots + \frac{a_n}{n+1}x^{n+1}$. Then $F(0) = 0$ and $F(1) = 0$ by hypothesis. Rolle gives $c \in (0, 1)$ with $F'(c) = 0$, i.e. $a_0 + a_1c + \cdots + a_nc^n = 0$. $\square$

The move — *invent an antiderivative so Rolle has something to bite on* — is one of the most frequently used tricks in olympiad analysis.

Example 2.4: A disguised Rolle via exponential multiplication. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable with $f(0) = f(1) = 0$. Prove there exists $c \in (0, 1)$ such that $f'(c) = f(c)$.

Solution. The conclusion $f'(c) - f(c) = 0$ is what we’d get by setting $g(x) = e^{-x}f(x)$ and demanding $g'(c) = 0$: indeed,

$$g'(x) = e^{-x}f'(x) - e^{-x}f(x) = e^{-x}(f'(x) - f(x)).$$

Since $e^{-x} > 0$, zeros of g' coincide with points where $f' = f$. Now g is continuous on $[0, 1]$, differentiable on $(0, 1)$, and $g(0) = f(0) = 0 = f(1) = g(1)$. Rolle gives $c \in (0, 1)$ with $g'(c) = 0$, i.e. $f'(c) = f(c)$. $\square$

When the conclusion is “ $f' = \lambda f$ ” or “ $f' = kf + \text{stuff}$,” the right move is almost always to multiply by $e^{-\lambda x}$ (or a suitable exponential) and apply Rolle. The pattern generalises: to handle $f'(c) + p(c)f(c) = 0$, try $g(x) = \mu(x)f(x)$ where $\mu(x) = e^{\int p}$ is the integrating factor from ODE theory.

Example 2.5: Rolle applied thrice. Let $f : [a, b] \rightarrow \mathbb{R}$ be three times differentiable, with $f(a) = f(b)$ and $f'(a) = f'(b) = 0$. Show there exists $c \in (a, b)$ with $f'''(c) = 0$.

Solution. Since $f(a) = f(b)$, Rolle gives $\alpha \in (a, b)$ with $f'(\alpha) = 0$. Now $f'(a) = f'(\alpha) = 0$: apply Rolle to f' on $[a, \alpha]$ to get $\beta_1 \in (a, \alpha)$ with $f''(\beta_1) = 0$. Similarly, $f'(\alpha) = f'(b) = 0$ gives $\beta_2 \in (\alpha, b)$ with $f''(\beta_2) = 0$. Finally, $f''(\beta_1) = f''(\beta_2) = 0$ lets us apply Rolle to f'' on $[\beta_1, \beta_2]$, producing $c \in (\beta_1, \beta_2) \subset (a, b)$ with $f'''(c) = 0$. $\square$

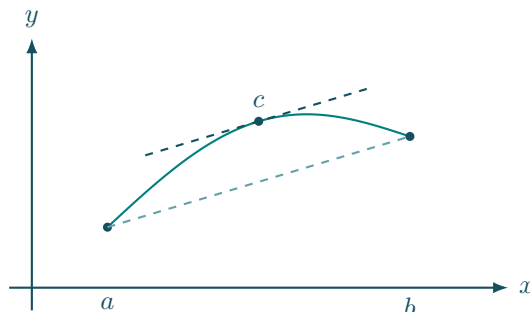
Repeated Rolle is a tree: each equal-value pair grows a new zero of the next derivative. This is the principle behind error-estimate proofs for polynomial interpolation (e.g., the Lagrange remainder theorem).

3 Lagrange’s Mean Value Theorem — the workhorse

Theorem (Lagrange’s Mean Value Theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Motivation. If a car averages 60 km/h over a one-hour trip, then at some instant its speedometer read exactly 60 km/h. Average rate of change is attained as an instantaneous rate somewhere.



Intuition. This is Rolle with training wheels off. Tilt the graph so the chord becomes horizontal — formally, apply Rolle to $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$ — and you recover the MVT. Every Lagrange proof is secretly a Rolle proof wearing a coat.

Which theorem to reach for?. Reach for **Lagrange** when you want to compare f at two points using info about f' in between. Typical signals: inequalities like $|f(x) - f(y)| \leq M|x - y|$, bounds on $f(b) - f(a)$ from bounds on f' , or deducing that f is monotonic/constant/Lipschitz.

Example 3.1: A logarithmic inequality. Show for all $0 < a < b$: $\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}$.

Solution. Apply MVT to $\ln$ on $[a, b]$: there is $c \in (a, b)$ with $\ln b - \ln a = \frac{b-a}{c}$. Since $a < c < b$, we get $\frac{1}{b} < \frac{1}{c} < \frac{1}{a}$, and multiplying by $(b-a) > 0$ gives the claim. $\square$

Inequalities with $\ln, e^x, \arctan, \sin$ are almost always MVT problems. Identify f , apply MVT on the right interval, bound $f'(c)$ using $a < c < b$.

Example 3.2: The Lipschitz punchline. If f is differentiable on $\mathbb{R}$ with $|f'| \leq M$, then $|f(x) - f(y)| \leq M|x - y|$.

Solution. MVT gives c between x and y with $f(x) - f(y) = f'(c)(x - y)$, so $|f(x) - f(y)| \leq M|x - y|$. $\square$

Global bounds on f' become global bounds on how fast f can change — the engine behind ODE uniqueness and contraction arguments.

Example 3.3: A classical bound for sin. Prove $|\sin x - \sin y| \leq |x - y|$.

Solution. MVT applied to $\sin$: $\sin x - \sin y = \cos c \cdot (x - y)$, and $|\cos c| \leq 1$. $\square$

Notice how this bypasses sum-to-product trig identities. Whenever a trig inequality has the shape “bound the difference,” MVT is the shortest route.

Example 3.4: An e^x inequality via MVT. Show $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, with equality iff $x = 0$.

Solution. Apply MVT to $f(t) = e^t$ on $[0, x]$ (for $x > 0$; the case $x < 0$ is symmetric and $x = 0$ is trivial): there is $c \in (0, x)$ with

$$e^x - e^0 = e^c(x - 0), \quad \text{i.e.} \quad e^x - 1 = x \cdot e^c.$$

Since $c > 0$, $e^c > 1$, so $e^x - 1 > x$, i.e. $e^x > 1 + x$. For $x < 0$, MVT on $[x, 0]$ gives $c \in (x, 0)$ with $1 - e^x = e^c(-x)$; now $e^c < 1$ and $-x > 0$, so $1 - e^x < -x$, i.e. $e^x > 1 + x$. $\square$

This is the canonical one-line proof of a genuinely useful inequality. The pattern — MVT with a monotone derivative gives a strict inequality of tangent-line type — reappears everywhere, from convexity proofs to Taylor-series bounds.

Example 3.5: A derivative-limit theorem. Let f be continuous on $[a, b]$ and differentiable on (a, b) . If $\lim_{x \rightarrow a^+} f'(x) = L$ exists, then f is differentiable at a from the right, and $f'(a^+) = L$.

Solution. For $h > 0$ small, apply MVT to f on $[a, a + h]$: there is $c_h \in (a, a + h)$ with

$$\frac{f(a + h) - f(a)}{h} = f'(c_h).$$

As $h \rightarrow 0^+$, $c_h \rightarrow a^+$ (it’s squeezed between a and $a + h$), so $f'(c_h) \rightarrow L$.

Hence $\lim_{h \rightarrow 0^+} \frac{f(a + h) - f(a)}{h} = L$, which is exactly $f'(a^+) = L$. $\square$

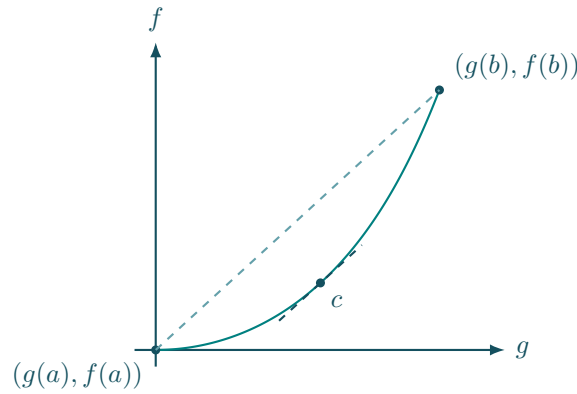
This explains a subtle point: if f' has a limit at a point, that limit automatically equals the derivative there — a derivative can’t have a removable or jump discontinuity. (You can see Darboux’s theorem smiling quietly in the background.)

4 Cauchy’s Mean Value Theorem — the comparator

Theorem (Cauchy’s Mean Value Theorem). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) , with $g'(x) \neq 0$ on (a, b) . Then there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Motivation. Imagine two runners whose positions at time t are $f(t)$ and $g(t)$. Cauchy says: at some moment c , the *ratio* of their instantaneous speeds equals the ratio of their total displacements.



Intuition. Setting $g(x) = x$ recovers Lagrange, so Cauchy is strictly more general. The proof applies Rolle to $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$.

Which theorem to reach for?. Reach for **Cauchy** when the quantity of interest is a *ratio*. The clearest signal: an expression $\frac{f(b)-f(a)}{g(b)-g(a)}$, or a $\frac{0}{0}$ limit (L'Hôpital is Cauchy in disguise).

Example 4.1: L'Hôpital demystified. Using Cauchy, prove: if $f(a) = g(a) = 0$, $g' \neq 0$ near a , and $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$.

Solution. Apply Cauchy to f, g on $[a, x]$: there is c_x between a and x with $\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c_x)}{g'(c_x)}$. Let $x \rightarrow a$; then $c_x \rightarrow a$ too, and the right side tends to L . $\square$

L'Hôpital's rule is literally Cauchy's MVT plus a limit.

Example 4.2: A ratio Lagrange can't handle. Find $c \in (1, 2)$ with $\frac{2^2 - 1^2}{2^3 - 1^3} = \frac{f'(c)}{g'(c)}$, where $f(x) = x^2, g(x) = x^3$.

Solution. LHS = $3/7$; RHS = $\frac{2c}{3c^2} = \frac{2}{3c}$. Setting equal: $c = \frac{14}{9}$, which lies in $(1, 2)$. $\square$

The question never involved the x -axis directly — just the pair (f, g) . That's Cauchy's natural habitat.

Example 4.3: An elegant comparison. For f differentiable on $[a, b]$ with $0 < a < b$, prove there exists $c \in (a, b)$ with

$$\frac{af(b) - bf(a)}{a - b} = f(c) - cf'(c).$$

Solution. Apply Cauchy to $F(x) = \frac{f(x)}{x}$ and $G(x) = \frac{1}{x}$ on $[a, b]$. Then $G'(x) = -1/x^2 \neq 0$. Compute:

$$\begin{aligned} \frac{F(b) - F(a)}{G(b) - G(a)} &= \frac{f(b)/b - f(a)/a}{1/b - 1/a} = \frac{af(b) - bf(a)}{a - b}, \\ \frac{F'(c)}{G'(c)} &= \frac{(cf'(c) - f(c))/c^2}{-1/c^2} = f(c) - cf'(c). \end{aligned}$$

Equating, we're done. $\square$

The art of Cauchy is choosing the right pair (F, G) . The asymmetric LHS practically whispers " $f(x)/x$ and $1/x$."

Example 4.4: Comparison of two growths. Let f, g be differentiable on $[0, 1]$ with $f(0) = g(0) = 0$, $f(1) = g(1) = 1$, and $g'(x) > 0$ on $(0, 1)$. Prove there exists $c \in (0, 1)$ with $f'(c) = g'(c)$.

Solution. By Cauchy, there is $c \in (0, 1)$ with $\frac{f(1) - f(0)}{g(1) - g(0)} = \frac{f'(c)}{g'(c)}$. The LHS equals $\frac{1 - 0}{1 - 0} = 1$, so $f'(c) = g'(c)$. $\square$

Two functions with the same endpoints must, at some interior point, have the same instantaneous rate. A tidy consequence: no matter how differently two paths from $(0, 0)$ to $(1, 1)$ curve, their slopes agree somewhere.

Example 4.5: A generalised mean value identity. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be differentiable with $g' \neq 0$. Show that for some $c \in (a, b)$,

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Solution. This is Cauchy's MVT rearranged. Define $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. Then

$$h(a) = [f(b) - f(a)]g(a) - [g(b) - g(a)]f(a) = f(b)g(a) - g(b)f(a),$$

$$h(b) = [f(b) - f(a)]g(b) - [g(b) - g(a)]f(b) = f(b)g(a) - g(b)f(a).$$

So $h(a) = h(b)$. Rolle gives $c \in (a, b)$ with $h'(c) = 0$, which is the claimed identity. $\square$

This is Cauchy's MVT in "product" form, which avoids the division and the $g' \neq 0$ hypothesis being needed for the *statement* (only for the ratio form). It's also the form that makes the proof a one-line Rolle application.

5 A cheat sheet

Which theorem, by given problem statement

Statement in the problem	Theorem to try first
Prove a root, fixed point, or meeting exists	IVT
Show two functions must agree somewhere	IVT on $h = f - g$
Discontinuous function has IVP	Darboux (if it's a derivative)
Count roots of f or f' ; $f(a) = f(b)$	Rolle
Weighted sum of coefficients equals zero	Rolle on the antiderivative
Conclusion involves $f^{(n)}(c) = 0$, several vanishing points	Repeated Rolle
Bound $f(b) - f(a)$ from bounds on f'	Lagrange
Prove f is Lipschitz, monotonic, or constant	Lagrange
Inequalities with $\ln, e^x, \arctan, \sin, \dots$	Lagrange
f' has a limit at a point $\Rightarrow$ derivative there	Lagrange (+ squeeze)
Compare the growth of two functions (a ratio)	Cauchy
$\frac{0}{0}$ indeterminate; L'Hôpital flavour	Cauchy
Two functions agree at two points, compare f', g'	Cauchy

Auxiliary-function transformations (the "coat" Rolle wears)

Most problems that ask for $f'(c) + (\text{expression in } f, c) = 0$ are Rolle problems disguised by an auxiliary function g . The art is recognising which g unlocks the conclusion.

Conclusion sought	Auxiliary function	Why it works
$f'(c) = 0$	$g(x) = f(x)$	Rolle directly.
$f'(c) = \lambda$	$g(x) = f(x) - \lambda x$	$g' = f' - \lambda$; chord becomes horizontal.
$f'(c) = \frac{f(b)-f(a)}{b-a}$	$g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$	Proves the MVT itself.
$f'(c) = f(c)$	$g(x) = e^{-x}f(x)$	$g' = e^{-x}(f' - f)$, so $g'(c) = 0 \Leftrightarrow f'(c) = f(c)$.
$f'(c) = \lambda f(c)$	$g(x) = e^{-\lambda x}f(x)$	$g' = e^{-\lambda x}(f' - \lambda f)$.
$f'(c) + p(c)f(c) = 0$	$g(x) = e^{\int p} \cdot f(x)$	Integrating factor from ODE theory.
$cf'(c) = f(c)$	$g(x) = \frac{f(x)}{x}$	$g' = \frac{xf' - f}{x^2}$.
$cf'(c) = kf(c)$	$g(x) = \frac{f(x)}{x^k}$	$g' = \frac{xf' - kf}{x^{k+1}}$.
$f'(c)g(c) = f(c)g'(c)$	$h(x) = \frac{f(x)}{g(x)}$	$h' = \frac{f'g - fg'}{g^2}$; or apply Cauchy.
$f'(c) = kf(c) + \ell(c)$	$g(x) = e^{-kx}(f(x) - \Phi(x))$	Φ a particular solution of the ODE.

The single most transferable move on this list is the one with e^{-x} (or $e^{-\lambda x}$). Whenever you see “ $f'(c) = \text{const} \cdot f(c)$ ” or related, multiply by the right exponential and Rolle or MVT is waiting.

6 Exercises

The exercises are loosely arranged by difficulty. Numbers 1–5 drill core moves; 6–10 require a bit of cleverness or choice; 11–15 are olympiad-register problems that combine techniques.

Problem 1. Prove that the equation $x \cdot 2^x = 1$ has exactly one positive real solution.

Problem 2. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous with $f(0) = f(1)$. Prove there exists $c \in [0, \frac{1}{2}]$ with $f(c) = f(c + \frac{1}{2})$.

Problem 3. Show that $3x^4 - 4x^3 - 12x^2 + C = 0$ has at most two distinct real roots in $[0, 2]$ for any real C .

Problem 4. Using the MVT, prove that for $x > 0$:

$$\frac{x}{1+x} < \ln(1+x) < x.$$

Problem 5. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and $\lim_{x \rightarrow \infty} f'(x) = L$. Prove $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = L$.

Problem 6. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, differentiable on $(0, 1)$, with $f(0) = 0, f(1) = 1$. Prove there exist distinct $c_1, c_2 \in (0, 1)$ with

$$\frac{1}{f'(c_1)} + \frac{1}{f'(c_2)} = 2.$$

Problem 7. Let $a_0, \dots, a_n \in \mathbb{R}$ with $\sum_{k=0}^n \frac{a_k}{2k+1} = 0$. Prove $a_0 + a_1x^2 + \dots + a_nx^{2n} = 0$ has a root in $(0, 1)$.

Problem 8. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable with $f(0) = 0$ and $f(1) = 1$. Prove there exists $c \in (0, 1)$ with $f'(c) = 2cf(c) + 1$. [*Hint: consider $g(x) = e^{-x^2}(f(x) + \text{something})$.*]

Problem 9. Give an example of a bounded function $g : [0, 1] \rightarrow \mathbb{R}$ that does *not* have the intermediate-value property, and conclude that g is not the derivative of any function on $[0, 1]$.

Problem 10. Let f be twice differentiable on $[a, b]$ with $f(a) = f(b) = 0$ and $f(c) > 0$ for some $c \in (a, b)$. Prove there exists $\xi \in (a, b)$ with $f''(\xi) < 0$.

Problem 11. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous on $[0, 1]$ and differentiable on $(0, 1)$, with $f(0) = 0$ and $|f'(x)| \leq |f(x)|$ for all $x \in (0, 1)$. Prove $f \equiv 0$ on $[0, 1]$. [*Hint: consider $e^{-x}|f(x)|$, or bootstrap from Lagrange.*]

Problem 12. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be differentiable with $f(0) = 0$ and f' monotonically increasing. Prove that $g(x) = \frac{f(x)}{x}$ is monotonically increasing on $(0, \infty)$. [*Hint: express g' using Cauchy's MVT.*]

Problem 13. Let f be differentiable on $[0, 1]$ with $f(0) = 0, f(1) = 1$. Prove that for every positive integer n , there exist distinct $c_1, \dots, c_n \in (0, 1)$ with

$$\sum_{k=1}^n \frac{1}{f'(c_k)} = n.$$

Problem 14. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Prove there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = \frac{f'(c)}{1} \cdot \frac{\ln b - \ln a}{\ln b - \ln a},$$

and further, for some $\xi \in (a, b)$ (assuming $0 < a < b$),

$$\frac{f(b) - f(a)}{\ln b - \ln a} = \xi f'(\xi).$$

Problem 15. Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable with $f'(a) = f'(b)$. Prove there exists $c \in (a, b)$ such that

$$f(c) - f(a) = (c - a)f'(c).$$

[*Hint: consider $h(x) = \frac{f(x) - f(a)}{x - a}$ extended to $x = a$ by $f'(a)$, and apply Rolle-style reasoning to h — or construct an appropriate auxiliary function and use the MVT.*]

Happy problem-solving!